

Huynh Le Dang Khoa

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Research Profile

Honors Computer Science graduate from Ho Chi Minh City University of Technology (HCMUT – VNU-HCM), with research experience spanning **computer vision, foundation models, and embodied AI**. My current research direction is centered on **vision–language–action (VLA) models and robotic learning**, particularly long-horizon manipulation, memory-augmented policies, learning from human video, and scalable evaluation of embodied agents.

My previous work studies reliable visual reasoning under sparse, corrupted, and distribution-shifted observations, including monocular metric depth, anomaly detection, and egocentric hand–object perception. I am interested in connecting these perceptual capabilities with robot policies that must **perceive reliably, retain information over time, and act under changing real-world conditions**.

Education

Ho Chi Minh City University of Technology (HCMUT – VNU-HCM) **Ho Chi Minh City, Vietnam**
B.Eng. Computer Science & Engineering — Honors Program 2022 – 2026

- **Academic Standing:** GPA: Excellent; Top 1% of Faculty. Program completed; official diploma expected November 2026.
- **Undergraduate Thesis:** Accuracy-aware and failure-resilient scheduling for concurrent vision inference; grade: **9.84/10**.
- **Research Interests:** Vision–language–action models, robotic learning, embodied perception, multimodal foundation models, long-horizon memory, and robot evaluation.

Research Experience

Eye Code Tech / ARENA Lab, Singapore Management University **Singapore / Remote**
Research Collaborator — Computer Vision & Robot Learning *Feb 2026 – Present*

- **Vision Foundation Models:** Research on reliable visual reasoning with frozen or pretrained foundation models, including monocular depth grounding, acquisition-robust anomaly detection, and egocentric manipulation perception.
- **Embodied AI:** Developing research directions that connect visual perception to robot learning, with emphasis on memory-augmented VLA policies, video-derived demonstrations, and long-horizon manipulation.
- **Research Collaboration:** Collaborative research involving ARENA Lab at Singapore Management University and Do Van Minh's group at SMART (Singapore–MIT Alliance for Research and Technology).

HPCC Lab, HCMUT **Ho Chi Minh City, Vietnam**
Research Assistant — GPU Inference, Scheduling & Systems *Jan 2025 – May 2026*

- **Efficient ML Inference:** Designed adaptive scheduling methods for concurrent heterogeneous vision workloads and evaluated accuracy–latency–throughput trade-offs under service-level constraints.
- **Thesis Results:** Achieved approximately **20% higher throughput** and **80% fewer SLA violations** than strong scheduling baselines on mixed ML inference workloads.
- **GPU Profiling:** Built reproducible inference benchmarks and profiled model latency, memory use, throughput, and scheduling behavior on modern NVIDIA GPUs.

Dien Toan Group

AI Engineer Intern

Ho Chi Minh City, Vietnam

May 2025 – Aug 2025

- **Retrieval-Augmented Generation:** Developed a document-understanding and RAG pipeline using multimodal language models, vector retrieval, PostgreSQL, MinIO, and Docker-based services.
- **Recognition:** Member of the team receiving the company’s **Most Outstanding Intern Team** award.

Publications & Manuscripts

Reliable Few-Point Metric Grounding for Monocular Depth Foundation Models. **ICLR 2027 submission, under review.** On NYUv2 with UniDepthV2, eight sparse metric observations reduce AbsRel from **0.0766** to **0.0540 (29.5% relative reduction)**; with 80% corrupted observations, AbsRel remains **0.0608**.

Acquisition-Aware Anomaly Detection. **Under review, Image and Computer Vision.** Introduces **ARAD**, a benchmark for anomaly detection under blur, motion, illumination, sensor noise, and mixed acquisition shifts, together with a training-free acquisition-aware detection framework.

Mutual Hand–Object Reasoning for Egocentric Manipulation Perception. **ICRA 2027 submission, under review.** Improves over CaRe-Ego by **6.93 mIoU points** on the EgoHOS out-of-distribution split and by **3.82 mIoU points** on mini-HOI4D in cross-dataset evaluation without fine-tuning.

An Intelligent Virtual Machine Consolidation Approach for Energy-Efficient Cloud Data Centers. **First author, ACOMP 2025.** [DOI](#)

Selected Research Projects

MemoryPi — Memory-Augmented Vision–Language–Action Model · Investigating explicit long-term memory for VLA policies. The policy stores and retrieves task-relevant information across subtasks so that actions can depend on observations or outcomes no longer visible in the current frame. Motivated by long-horizon failure modes observed in RoboMME-style manipulation tasks.

Video-to-Simulation for Robot Learning · Building a pipeline that uses computer-vision models to recover object geometry, hand–object interaction, and demonstrations from monocular human video, then retargets the reconstructed interaction into a robot simulation environment for learning and evaluation.

Generative Evaluation for VLA Policies · Exploring action-conditioned robot video generation and learned world models as a complementary evaluation signal for scenes that are difficult to reproduce with manually authored 3D assets. Current focus is understanding when predicted visual futures are sufficiently reliable to evaluate robot behavior.

Technical Skills

Programming: Python, C/C++, Java, Go, SQL, Linux, Git.

Machine Learning: PyTorch, OpenCV, CUDA; vision and multimodal foundation models; object detection and segmentation; monocular depth; VLA models; reinforcement learning; RAG and agentic AI.

Robotics & Simulation: Isaac Lab, robot simulation, manipulation evaluation, video-to-simulation, vision-based robot learning.

Compute: Access to NVIDIA **H100**, RTX **A5000**, and experience profiling and optimizing GPU inference workloads.

Awards & Additional Information

Awards: Most Outstanding Intern Team — Dien Toan Group; 2nd Place — Innostart Entrepreneurship Competition; Top 4 — UTE Junior Startup.

Languages: Vietnamese (Native); English (Professional Working Proficiency).